

GENERAL SPECIFICATION G5.3

Electrical General Provisions

CONTENTS

G5.3 Electrical General Provisions1

 G5.3.1 General.....1

 G5.3.2 Products3

 G5.3.3 Execution.....5

G5.3 ELECTRICAL GENERAL PROVISIONS

G5.3.1 General

(a) Introduction

- (i) General Specifications G5, Particular Specification P5 and Drawings for Electrical Works covers the work necessary for the complete electrical system that is fixed in position, connected, tested, safe and in fully functional working order. Supply, deliver, install, test and commission materials and equipment in accordance with General and Particular Specification, Drawings, and all other documents with respect to this Contract.
- (ii) The work shall include provision of all labour and all materials necessary to form a complete installation and such tests, adjustments and commissioning as are prescribed in subsequent sections, compatible to the Plant's functional & operational system requirements and as may otherwise be required to give an effective working installation to the satisfaction of the S.O..
- (iii) Supply and install all incidental items not shown or specified, but which are required by good practice, safe and reliable operation and maintenance to provide complete functional systems.
- (iv) The Contractor shall ensure that all requirements are strictly adhered to when carrying out work required by the Contract Documents (General and Particular Specifications, and Drawings), unless otherwise instructed by the S.O..

(b) Submittals

- (i) Submittals shall be made as required in General Specification G1.2 - Information Management.

(c) Design Requirements

- (i) All equipment anchoring and mounting shall be in accordance with the manufacturer's requirements unless otherwise indicated.
- (ii) Provide materials and equipment installed outdoors, in direct sunlight, capable of continuous operation at their specified ratings within an ambient temperature range up to 40 degrees C, and humidity up to 95%. Equipment, Instruments, control devices, and electrical panels, installed outdoor shall be protected with proper shelter or shade of appropriate size and dimension to protect the equipment from rain or shine.
- (iii) All cable trays, cable ladders, trunking, conduits, supports, glands, hangers, brackets and fixings shall be constructed from Type 316 Stainless Steel materials due to the generally corrosive environment of the Plant. The Contractor shall verify the adequacy of all cable trays and cable ladders and based on actual site applications & suitability prior to procurement. Increase sizes where required to allow a minimum of 25 percent spare space for installation of future cables and maintenance.

- (d) Electrical Coordination
 - (i) Work provided outside of this Contract: Liaise and co-ordinate the electrical works being installed under this Contract with the electrical Works being installed under other Contracts.
 - (ii) Co-ordination between Trades: Liaise and co-ordinate the electrical interface requirements for all equipment and construction which affect the work covered under General Specifications G5 and Particular Specification P5.
- (e) Standards, Codes and Regulations
 - (i) Perform all work; supply and install all materials and equipment in full accordance with, but not limited to, the latest applicable rules, regulations, requirements, national and international codes, as listed in Specification G5.2 – Standards, Codes and Regulations.
 - (ii) Conflicts, if any, that may exist between the above reference standards, codes and regulations will be resolved at the discretion of the S.O..
- (f) Operation and Maintenance Manuals
 - (i) Provide Operations and Maintenance Manuals in accordance with the General Specification G1.2 - Information Management. In addition to the requirements of the General Specifications, submit the following detailed information:
 - 1) Index of all equipment manufacturers listing current names, addresses, and telephone numbers of those who should be contacted for service, information and assistance;
 - 2) Equipment datasheets, FAT test reports, SAT test reports, SIT (System Integration Tests), SAT (Site Acceptance Tests), performance test reports, configuration files, troubleshooting charts / diagrams, parameter setting of motor starters e.g. VFDs, soft starters, controller unit details, PLC programme details and all necessary and essential information for the effective operation etc. shall be submitted along with the O&M manuals;
 - 3) As-built circuit connection diagrams, control schematics of motor starters submit in electronic files;
 - 4) As-built circuit identification schedule, submit electronic files;
 - 5) Provide as-built single-line diagrams for all electrical equipment;
 - 6) An overall single line diagram with design calculations & protection settings for all equipment contained in each electrical room shall be embossed, endorsed by Electrical PE (Singapore), framed and hanged securely within the vicinity of the main circuit breaker in that electrical room;
 - 7) A copy of each distribution panel with design calculations & protection settings duly endorsed by Electrical PE (Singapore) shall be secured near it; and
 - 8) Provide suitable pocket with glass window and lock set to secure the diagram and for easy replacement of diagram.

- (g) Contract Drawings
 - (i) Electrical Contract drawings are only conceptual representation of the electrical system proposed. Contractor shall fully design, provide detailed calculations and develop all electrical drawings in accordance with local codes, national and international standards, and endorsed by LEW with appropriate Engineer grade.
 - (ii) Electrical plan drawings show only general tentative locations of equipment, devices, and raceway, unless specifically dimensioned. The Contractor shall be responsible for finalizing the location of equipment based on equipment information from the manufacturer (e.g. dimensions, clearances, etc.), and the proper routing of cables and raceway, subject to the approval of the S.O.. The raceway route shown on the drawings are only indicative for allocation of space. Contractor shall design and size cable raceways as per finalized cable routes.
 - (iii) All small equipment and devices (e.g. LCS, LCP, termination boxes and accessories, junction boxes, fuse holder etc.) are not shown on the drawings and are included in the scope of works under this contract.
 - (iv) Contractor's specific attention is drawn that all control, interlock and data cables are not shown on the drawings or cable schedule for all high voltage and low voltage equipment and are included in the scope of works under this contract. Contractor shall design and develop control schematics, wiring diagram and termination drawings, cable routing, power and control cable schedules and obtain approval from S.O..

G5.3.2 Products

- (a) General
 - (i) Unless otherwise indicated, provide all first quality, new materials and equipment, free from any defects, in new condition, and suitable for the space provided from reputable manufacturers.
 - (ii) Where two or more units of the same class of material or equipment are required, provide products of a same manufacturer. Component parts of materials or equipment need not be products of the same manufacturer.
 - (iii) Unless otherwise specified, all works and materials shall comply with the relevant Singapore Standards as laid down by SPRING Singapore. Where this is not possible, the relevant IEC or British standards or other S.O.'s approved standard shall apply.
- (b) Standard Products
 - (i) Unless otherwise indicated, provide materials and equipment that are the standard products of manufacturers regularly engaged in the production of such materials and equipment. Provide the manufacturer's latest standard design that conforms to General Specification G5.
 - (ii) The products offered by manufacturers shall not be obsolete or slated to be obsolete and spare parts shall be available for the life of the equipment for the operation and maintenance of the product.

- (c) Equipment Finish
 - (i) Provide materials and equipment with manufacturer's standard finish and colour, except where specific colour is indicated.
 - (ii) If manufacturer has no standard colour, finish equipment in accordance with light grey colour RAL 7035 unless otherwise stated in the equipment Specifications.
- (d) Hazardous Areas
 - (i) Hazardous Areas are areas where explosive gas/air/dust/fibre mixtures are likely to exist. These are defined in the Hazardous Area Classification table, which is in Particular Specification P5. The Contractor shall:
 - 1) Provide materials and equipment acceptable to the regulatory authority having jurisdiction for the level of hazard indicated, and in accordance with IEC 60079;
 - 2) All hazardous area classified apparatus shall have either IEC-Ex certification;
 - 3) All hazardous area classified equipment / devices shall be certified for use the zone and conditions stipulated in the Hazardous Area Classification table;
 - 4) All equipment for hazardous areas shall be supplied with an IEC-Ex certificate of conformity. Equipment for Hazardous areas shall be provided with the detailed information name plate for all classes, supplied and installed. For each model supplied, Tag numbers and location shall be indicated on the panel. One certificate for each model used shall be sufficient. Original certificates shall be arranged in a dedicated file. Copies shall be included in the Operation & Maintenance Manuals;
 - 5) "Ex" warning signs shall be installed at hazardous area; and
 - 6) Ex proof equipment shall have warning sign stating 'WARNING- DO NOT OPEN UNTIL POWERED OFF'.
- (e) Shipping, Storage, Handling and Protection
 - (i) Following installation, protect materials and equipment from corrosion, physical damage, and the effects of moisture on insulation and contact surfaces. During delivery and transportation for the equipment intended for indoor installation in areas where it is subject to dampness, moisture, dirt, or other adverse atmosphere until completion of construction, ensure that adequate protection from these atmospheres is provided that is acceptable to S.O.. Cap conduit runs during construction with manufactured seals. Keep openings in boxes or equipment closed during construction. Energise all space heaters supplied with equipment. VFDs, Soft Starters or equipment with electronic cards, PLC or any specialised equipment shall be handled properly in accordance with manufacturer's recommendations and requirement.

G5.3.3 Execution

- (a) General
 - (i) Liaise and co-ordinate electrical work with S.O. and work of other trades to avoid conflicts, errors, delays and unnecessary interference with others during construction.
 - (ii) Check the approximate locations of light fixtures, electrical sockets, equipment, and other electrical system components shown on Drawings for conflicts with openings, structural members and components of other systems and equipment having fixed locations. In the event of conflicts, notify S.O. in writing. S.O.'s decision shall govern. Make modifications and changes necessary to eliminate conflicts.
 - (iii) Electrical equipment locations shown on the drawings are only indicative.
 - (iv) Electrical installation in areas classified as finished, indoor dry shall be concealed (embedded in concrete walls). All cables shall be installed in conduit in the walls. Cable tray or trunking may be installed above the ceiling.
- (b) Hazardous Areas
 - (i) Install all materials and equipment in hazardous areas in a manner acceptable to the regulatory authority having jurisdiction and in accordance with IEC 60079.
- (c) Load Balance
 - (i) Balance electrical load between phases as nearly as possible on switchgears, switchboards, distribution boards, motor control centres and other equipment where balancing is required.
 - (ii) When loads must be reconnected to different circuits to balance phase loads, maintain accurate record of changes made, and provide circuit directory that lists final circuit arrangement.
- (d) Removal or Relocation of Materials and Equipment
 - (i) Where materials or equipment are removed or relocated, remove all materials no longer used such as studs, straps, conduits, and wires. Remove or cut off concealed or embedded conduits, boxes, or other materials to a point at least 25 mm below the final finished surface.
 - (ii) Repair affected surfaces to conform to the type, quality, and finish of the surrounding surface in a neat and workmanlike manner. Follow any specific instructions given in the General Specifications, utilising skilled craftsmen of the trades involved.
- (e) Strutting and Reinstatement
 - (i) Do not cut or notch any structural member or building surface without written approval of S.O.. Carefully carry out any cutting, channelling, chasing, or drilling of floors, walls, partitions, ceilings, paving or other surfaces required for the installation, support or anchorage of conduits, raceways or other electrical materials and equipment. Following such work, restore surfaces neatly to original condition. Use skilled craftsmen of the trades involved.

- (f) Cleaning and Touch-Up Painting
 - (i) Keep the premises free from accumulation of waste material or rubbish. Upon completion of work, remove all materials, scraps, and debris from premises and from interior and exterior of all devices and equipment. Electrical switchrooms, XLV rooms, battery and UPS rooms shall be cleaned before and after installing the equipment. Touch up scratches, scrapes, or chips in interior and exterior surfaces of devices and equipment with finishes matching as nearly as possible the type, colour, consistency, and type of surface of the original finish. If extensive damage is done to equipment paint surfaces, refinish the entire equipment in a manner that provides a finish equal to or better than the factory finish, that meets the requirements of the General and Particular Specifications, and that is acceptable to S.O..
- (g) Hazardous Area Classification (HAC)
 - (i) Hazardous Areas are areas where explosive gas/air/dust/fibre mixtures are likely to exist.
 - (ii) Areas designated as hazardous shall be shown in the “Hazardous Area Zone Drawings” which shall be developed by the Contractor during detailed design stage. These drawings shall be based on the Hazardous Area Classification (HAC) document included as a supplement to the Particular Specification P5. This HAC is based on the design contained in the tender documents. Review, verification and submission shall be completed by the Contractor to assess any adjustments needed due to the equipment selected by the Contractor.

END OF SECTION